

# Sampling, FFT, and the Pre-Lab 1 notebook

Fall 2026 · Prof. Ehsan Roohi

**Lecture and guided Jupyter walkthrough**

# Learning goals

- Explain sample rate, sample interval, record length, and Nyquist frequency
- Run prepared analysis cells in a Jupyter notebook
- Interpret time histories, RMS values, and one-sided FFT spectra
- Recognize aliasing before collecting Lab 1 sound data

# Connection to Lab 1

## PRE-LAB

A known voltage signal contains two pure tones.

Amplitude describes signal size; frequency describes how fast it repeats; phase sets its starting position.

## LABORATORY

A microphone converts changing sound pressure into changing voltage.

Moku:Go stores voltage samples. The waveform shows change in time; the spectrum shows the frequencies present.

# Jupyter notebook structure

## MARKDOWN CELLS

Instructions, equations, and response areas.

Double-click to edit. Use Shift+Enter to render and move forward.

## CODE CELLS

Python commands that define variables, calculate results, and make figures.

Run from top to bottom with Shift+Enter.

# Opening and saving the notebook

- Download the .ipynb file from Canvas or open it from the course website in Colab
- Save a personal copy before editing
- Run cells from top to bottom
- Keep code outputs and figures in the submitted file

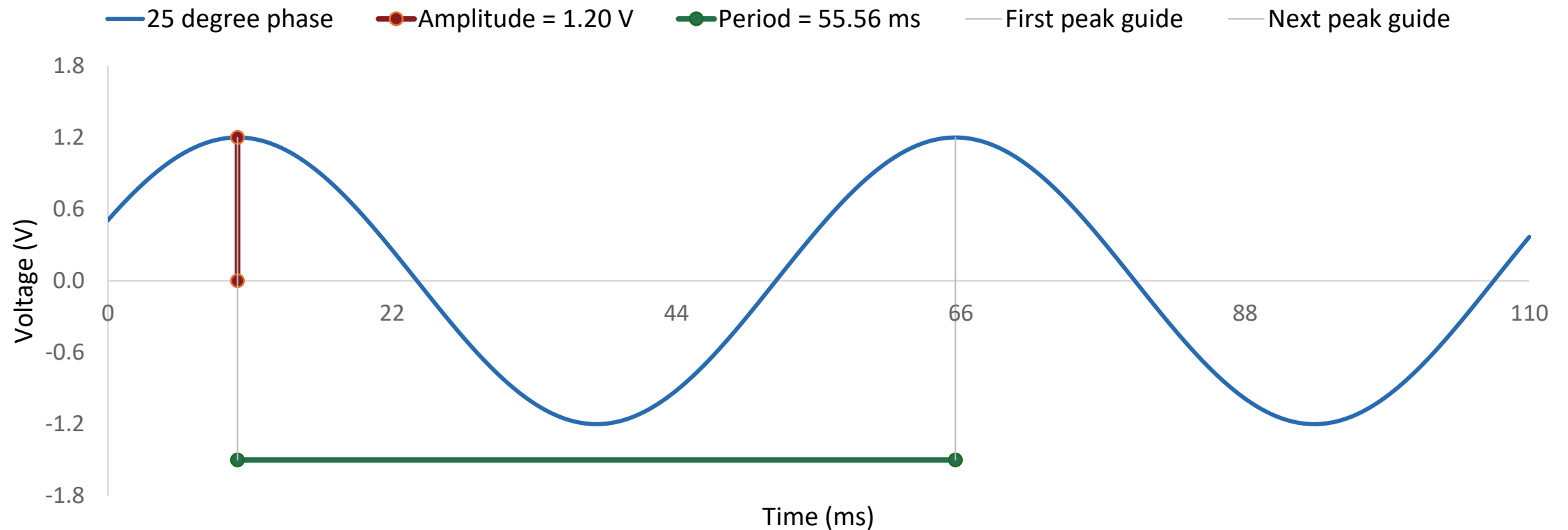
# Execution order matters

```
import numpy as np
A = 1.20
print(A)

# A later cell can use A only after this cell runs.
rms_component = A / np.sqrt(2)
print(rms_component)
```

A restart clears stored variables. Run All rebuilds the notebook state from the first cell.

# A sinusoid: what the three parameters mean



**$A_1 = 1.20$  V from zero to peak.  $T_1 = 55.56$  ms between consecutive peaks.**

$$x_1(t) = A_1 \sin(2\pi f_1 t + \phi_1), \text{ with } f_1 = 18 \text{ Hz and } \phi_1 = 25^\circ$$

# The new two-tone signal

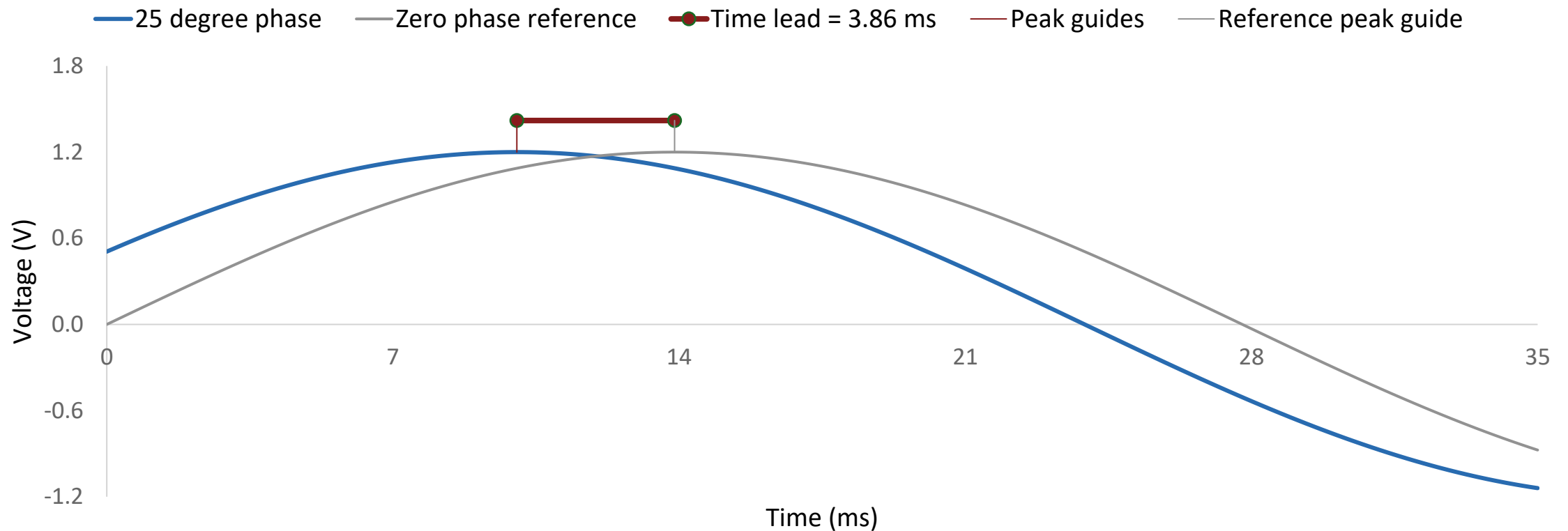
```
A1, f1, phi1_deg = 1.20, 18.0, 25.0
A2, f2, phi2_deg = 0.45, 42.0, -15.0

def signal(t):
    return (A1*np.sin(2*np.pi*f1*t + np.deg2rad(phi1_deg))
            + A2*np.cos(2*np.pi*f2*t + np.deg2rad(phi2_deg)))
```

Subscript 1 means tone 1; subscript 2 means tone 2. A is in volts, f is in hertz, and phi\_deg is in degrees.



# Phase relative to a zero-phase sine wave



**$\phi_1 = +25^\circ$ : the blue wave reaches its peak 3.86 ms earlier**

Time lead =  $\phi_1 / (360^\circ f_1)$ . Positive phase shifts a sine wave to the left.

# Mean and RMS

## MEAN

Average signed level:

$$\text{mean} = (1/N) \sum x[n]$$

N is the sample count;  $x[n]$  is sample n.

The continuous one-second signal has zero mean.

## RMS

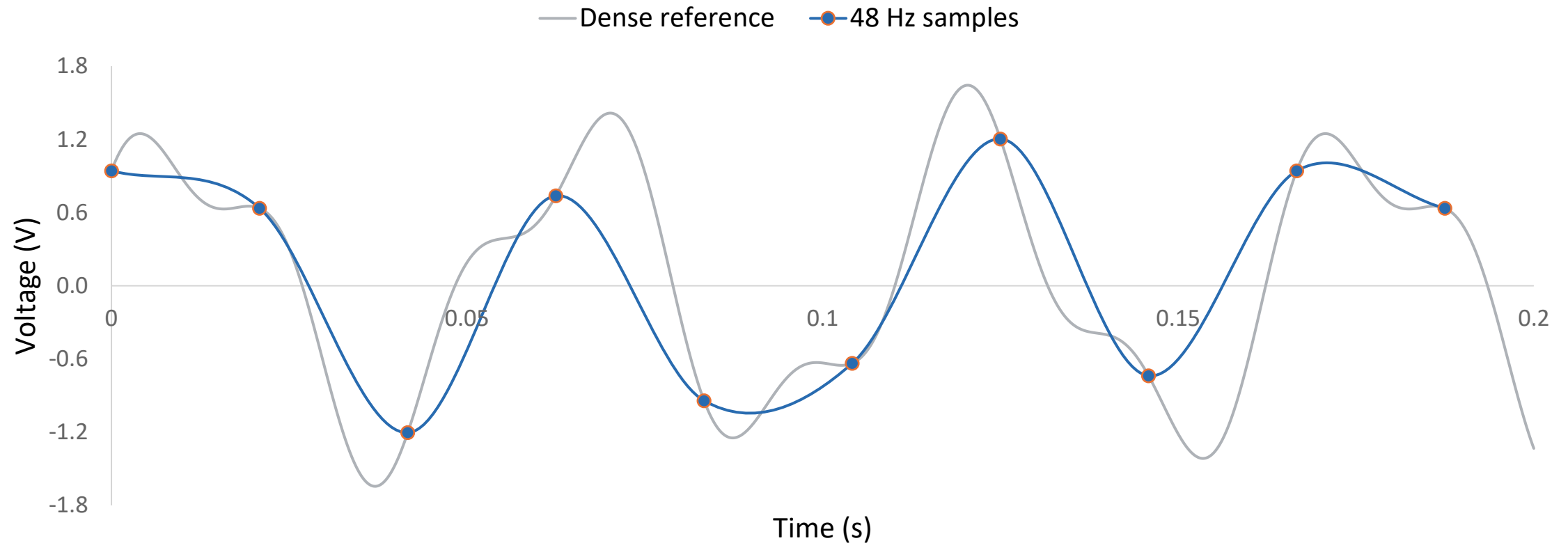
Typical signal size, independent of sign:

$$\text{RMS} = \sqrt{(1/N) \sum x[n]^2}$$

For these two distinct tones over complete cycles:

$$\text{RMS} = \sqrt{(A_1^2 + A_2^2)/2} = 0.9062 \text{ V}$$

# Samples and the underlying waveform



Markers are simulated samples of the known signal. Connecting lines add no measurements.

# Signal frequency and sampling rate

## THE WAVE: $f_{\text{signal}}$

$f_{\text{signal}}$  counts complete cycles per second.

Unit: hertz (Hz).

Our signal has two components:

$f_1 = 18 \text{ Hz}$  and  $f_2 = 42 \text{ Hz}$ .

## THE RECORDER: $f_s$

$f_s$  counts stored samples per second.

Unit: samples/s (often written Hz).

We choose this setting on the recorder.

Changing  $f_s$  changes the sample times.

# How many samples describe one cycle?

## DURATION OF ONE CYCLE

Signal period:  $T_{\text{signal}} = 1/f_{\text{signal}}$ .

For 42 Hz:  $T_{\text{signal}} = 23.81 \text{ ms}$ .

Sample interval:  $\Delta t = 1/f_s$ .

For 126 samples/s:  $\Delta t = 7.94 \text{ ms}$ .

## SAMPLES PER CYCLE

Samples/cycle =  $T_{\text{signal}} / \Delta t$

=  $f_s / f_{\text{signal}}$

$126 / 42 = 3$  samples per cycle.

$336 / 42 = 8$  samples per cycle.

# The Nyquist condition

## THE SIGNAL REQUIREMENT

$f_{\text{max}}$  is the highest frequency present.

For our two tones:  $f_{\text{max}} = 42$  Hz.

To avoid aliasing in this signal:

$f_s > 2 f_{\text{max}} = 84$  samples/s.

## THE RECORDER LIMIT

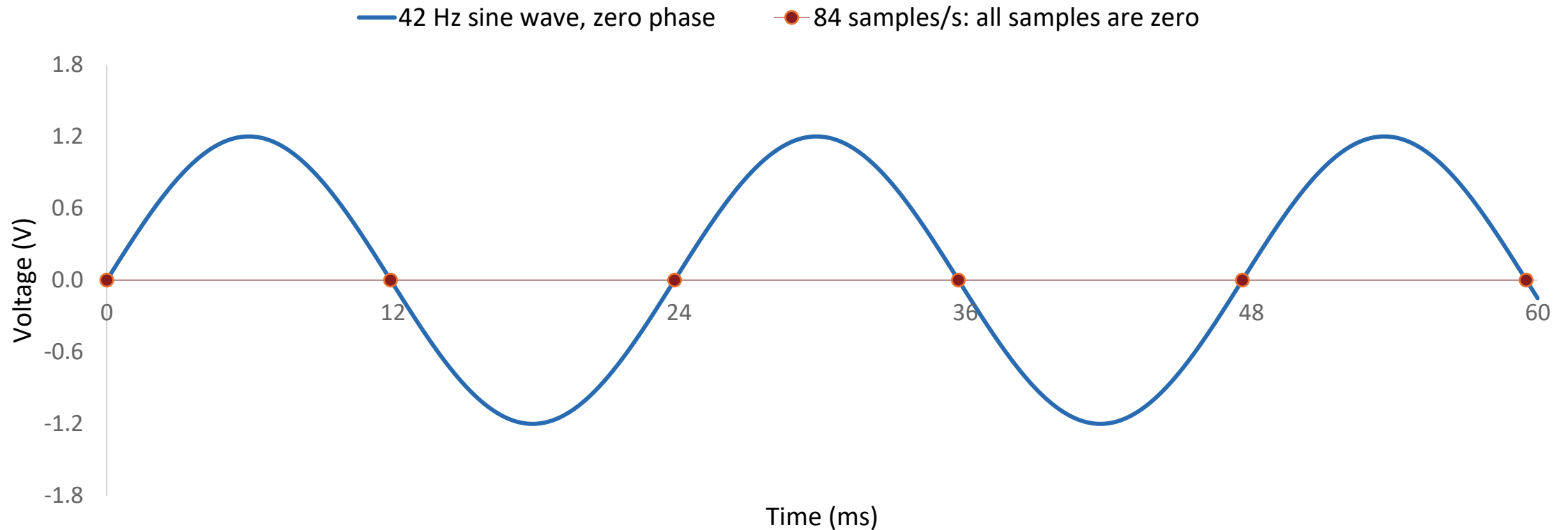
Nyquist frequency:  $f_N = f_s / 2$ .

At  $f_s = 126$  samples/s:  $f_N = 63$  Hz.

Both 18 Hz and 42 Hz are below 63 Hz.

The same condition is  $f_{\text{max}} < f_N$ .

## Exactly two samples per cycle can fail



**At  $f_s = 2 f_{\text{signal}}$ , this starting phase makes every reading zero.**

Use  $f_s > 2 f_{\text{max}}$  and leave margin for real measurements.

## Checking the four notebook sample rates

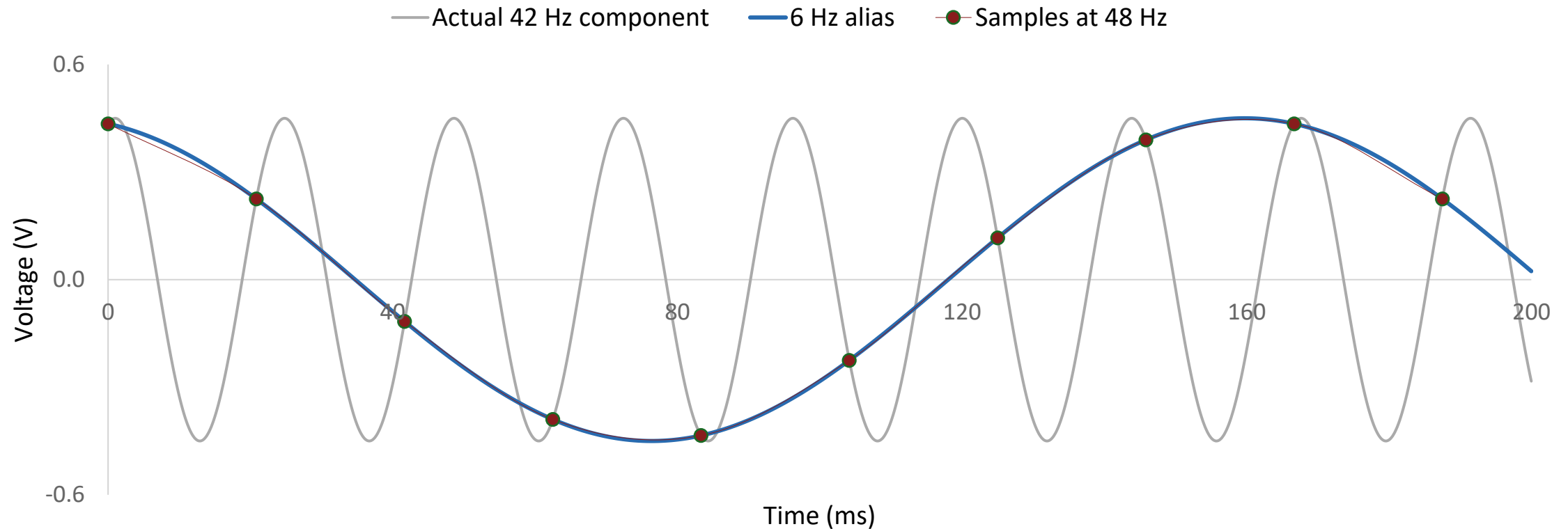
| $f_s$ (samples/s) | $f_N = f_s/2$ (Hz) | Samples / 42 Hz cycle | Both tones below $f_N$ ? |
|-------------------|--------------------|-----------------------|--------------------------|
| 336               | 168                | 8.00                  | Yes                      |
| 126               | 63                 | 3.00                  | Yes                      |
| 72                | 36                 | 1.71                  | No: 42 Hz aliases        |
| 48                | 24                 | 1.14                  | No: 42 Hz aliases        |

**The smaller 42 Hz component still sets  $f_{\max} = 42$  Hz.**

A small amplitude does not remove the need to sample that frequency correctly.



# Aliasing: two waves fit the same samples



**Both waves give the same sample values at  $f_s = 48$  Hz**

42 Hz folds to  $|42 - 48| = 6$  Hz. The signal itself has not changed.

# Predicting an alias frequency

## FIRST CHECK THE LIMIT

Compute  $f_N = f_s / 2$ .

If  $0 \leq f_{\text{signal}} < f_N$ , keep its frequency.

For  $f_N < f_{\text{signal}} < f_s$ :

$f_{\text{alias}} = f_s - f_{\text{signal}}$ .

## EXAMPLES FROM THIS NOTEBOOK

42 Hz at  $f_s = 72$ :  $f_N = 36$  Hz.

$f_{\text{alias}} = 72 - 42 = 30$  Hz.

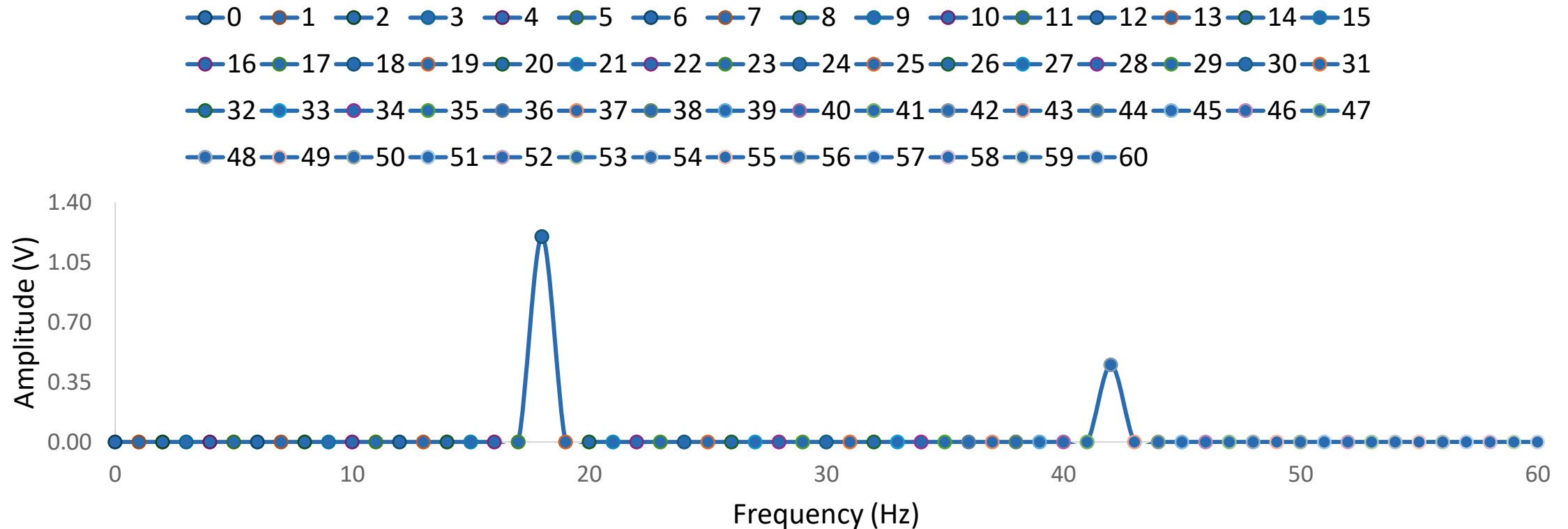
42 Hz at  $f_s = 48$ :  $f_N = 24$  Hz.

$f_{\text{alias}} = 48 - 42 = 6$  Hz.

# Preventing aliasing during acquisition

- Estimate the highest frequency in the input, including unwanted high-frequency content.
- Choose a sampling rate above twice that frequency, with practical margin.
- An analog low-pass filter reduces unwanted frequencies before sampling.
- After aliasing occurs, an FFT or a smoother plot cannot restore the lost information.

# A waveform and a spectrum answer different questions



**18 Hz peak: 1.20 V. 42 Hz peak: 0.45 V. Same samples, another view.**

Calculated from 336 samples in 1 s. Both tones complete whole cycles; no taper in this example.

# Reading an FFT peak

## HORIZONTAL POSITION

The x-axis is frequency in Hz.

A peak at 18 Hz means 18 cycles/s.

This axis is not time.

It is not the recorder sampling rate.

## VERTICAL HEIGHT

Here the y-axis is amplitude in volts.

The 18 Hz peak is about 1.20 V.

This is peak amplitude, not RMS.

Its single-tone RMS is  $1.20/\sqrt{2}$  V.

# Correct RMS does not guarantee correct frequency

## RMS SUMMARIZES SIGNAL SIZE

Square each value, average, then take a square root.

It compresses the record into one number.

It does not say how fast the wave repeats.

## THE NOTEBOOK COMPARISON

An aliased record can still have nearly the expected RMS.

Check the waveform and the FFT peaks as well as the mean and RMS.

# Record duration and FFT frequency bins

## THREE ACQUISITION QUANTITIES

$N$  = number of stored samples.

$f_s$  = samples per second.

$T_{\text{record}} = N / f_s$ , in seconds.

$T_{\text{record}}$  is not the period of one wave.

## THE FFT FREQUENCY GRID

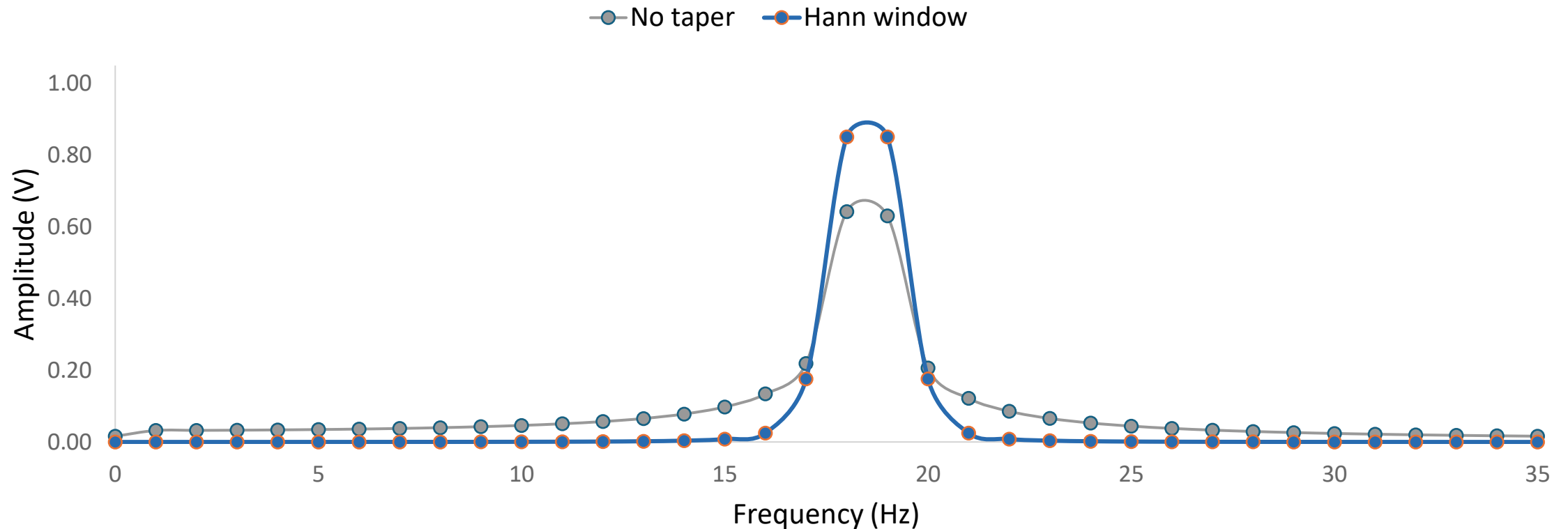
A bin is one frequency checked by the FFT.

Bin spacing:  $\Delta f = f_s / N = 1/T_{\text{record}}$ .

$T_{\text{record}} = 1 \text{ s}$  gives bins 1 Hz apart.

$T_{\text{record}} = 0.25 \text{ s}$  gives bins 4 Hz apart.

# Why a peak can spread over several bins



**One 18.5 Hz tone lies between 1 Hz bins. Neighboring bins are not extra tones.**

Same 1 s record in both curves. The Hann window reduces distant leakage but widens the peak.



# What the Hann window changes

## SMALLER VALUES NEAR THE ENDS

A window multiplies samples by weights.

The Hann weights taper toward zero at both ends.

This reduces distant leakage around a peak.

## A TRADEOFF

The main peak becomes wider.

Amplitude needs a scaling correction.

The provided code applies the correction.

Windowing does not fix aliasing.

# Sample rate and duration solve different problems

## HIGHER $f_s$ AT FIXED DURATION

More readings in each second.

Higher Nyquist frequency  $f_N = f_s/2$ .

The FFT-bin spacing stays  $1/T_{\text{record}}$ .

The physical tone frequency stays the same.

## LONGER $T_{\text{record}}$ AT FIXED $f_s$

More cycles of the signal are observed.

Finer FFT-bin spacing  $\Delta f = 1/T_{\text{record}}$ .

The Nyquist frequency stays the same.

The window also affects peak separation.

# Choosing settings for a new measurement

## ILLUSTRATIVE DESIGN TARGET

Suppose frequencies extend up to 60 Hz.

We want FFT bins no more than 0.5 Hz apart.

Choose  $f_s > 120$  samples/s.

Choose  $T_{\text{record}} \geq 2$  seconds.

## ONE POSSIBLE CHOICE

$f_s = 300$  samples/s,  $T_{\text{record}} = 2$  s.

$N = 600$  samples.

$f_N = 150$  Hz and  $\Delta f = 0.5$  Hz.

Check filtering, noise and device limits.

## Using the prepared notebook

- Open the course Colab link, save a personal copy, and run cells from top to bottom.
- Read fs and duration before interpreting each figure. The full code is already provided.
- Predict peaks first, then compare them with the printed output and plots.
- Enter calculations and explanations in the response cells. No code writing is required.

# What a good analysis answer includes

- Name the case: sampling rate, record duration and signal component.
- State a prediction with a calculation, including units.
- Report the observed value and compare it with the prediction.
- Explain the cause: adequate sampling, aliasing, leakage or limited record duration.

# Check your understanding

## QUESTIONS

1. Does raising  $f_s$  change  $f_1$  or  $f_2$ ?
2. Which quantity equals  $f_s/2$ ?
3. Can a longer record repair aliasing?
4. What does a peak location tell you?

## ANSWERS

1. No, it changes the sampling times.
2. The Nyquist frequency  $f_N$ .
3. No, choose adequate sampling first.
4. The frequency represented in the data.

# Pre-Lab 1 submission

## BEFORE SUBMITTING

Run all prepared cells.

Keep all figures visible.

Complete every response section.

Export to HTML or PDF for Canvas.

## DEADLINE

Normally posted Monday before the lab.

Submit before your own section begins.

Late pre-labs are not accepted.